

Programming 1: Lab 3 : Operators, Expressions and Conditions

Write the python code for the following questions. Handle all the valid and invalid test cases. Write down relevant comments in your code:

1. Write a program that prompts the user to enter the centre of a circle (x1, y1), its radius r, and a second point (x2, y2), and displays whether the second point lies inside, on the boundary of, or outside the circle. The formula for computing the distance between two points is
$$\text{distance} = \sqrt{(x2 - x1)^2 + (y2 - y1)^2}$$
2. Write a program that prompts the user to enter the side of a regular pentagon and displays its area. The formula for computing the area of a pentagon is $\text{Area} = (5 * s^2) / (4 * \tan(\pi / 5))$, where s is the length of a side.
3. Take 3 angles as input, and find whether they can form the angles of a triangle or not. If they can, further classify the triangle as acute-angled, right-angled or obtuse-angled. Consider invalid cases also.
4. Take a 4 digit number as input and find the sum of its first two digits and the sum of its last two digits separately. Also, check if the two sums are equal or not.
5. Take a 5 digit number as input and print the largest digit of the number. Do not use any in-built functions and do not use loops. Also print the position of that digit counted from the left. If the largest digit occurs more than once, print the position of its first occurrence.
6. Rotate the values of three integer variables cyclically, so that the value of a moves to b, the value of b moves to c, and the value of c moves to a, without using a fourth variable or multiple assignment operation.
7. Take a 3 digit number as input. Check if it is a Harshad number or not, i.e. whether the number is exactly divisible by the sum of its digits. E.g. $1 + 5 + 3 = 9$ and $153 / 9 = 17$, so 153 is a Harshad number.
8. Suppose a water tank initially holds V litres of water. Every day 5% of the water present in the tank evaporates, and then exactly 10 litres of fresh water is added to it. Therefore, the daily retention rate is $1 - 0.05 = 0.95$. After the first day, the quantity in the tank becomes
$$V * 0.95 + 10$$

After the second day, the quantity in the tank becomes
$$(V * 0.95 + 10) * 0.95 + 10$$

After the third day, the quantity in the tank becomes
$$((V * 0.95 + 10) * 0.95 + 10) * 0.95 + 10$$

and so on.
Write a program that prompts the user to enter the initial quantity of water and the number of days (N) and displays the quantity of water in the tank after the Nth day. Check for valid and invalid cases. Do not use loops.
9. Enter the coefficients of two straight lines $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ and display their point of intersection. Handle all the cases for invalid input as well as the cases where the two lines are parallel or coincident, and display the solution till exactly 2 decimal places.